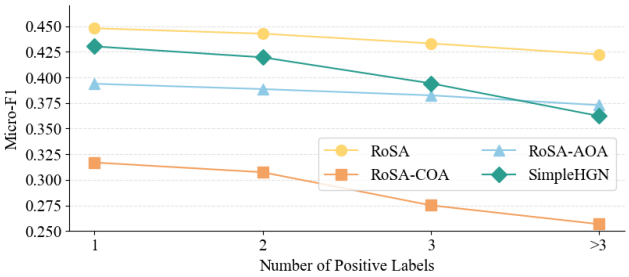

Figure 1. Performance under stratified evaluation by the number of positive labels L.

AOA contribution is defined as:

$$\rho(v) = \frac{\left\| g_v^{(2)} \odot h_{\mathrm{AOA}}(v) \right\|_2}{\left\| g_v^{(1)} \odot h_{\mathrm{COA}}(v) \right\|_2 + \left\| g_v^{(2)} \odot h_{\mathrm{AOA}}(v) \right\|_2} \quad (1)$$

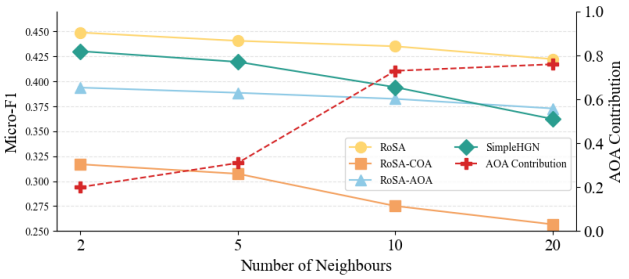

Figure 2. Secondary-neighbor stress test. Model performance: left y-axis; AOA contribution: right y-axis.

Table 1. Meta-path study on the CITE dataset.

Meta-path	Micro-F1	Macro-F1	Sample-F1
Paper-Keywords	0.4006	0.0120	0.3994
Paper-Author	0.3836	0.0091	0.3640
Paper-Substrate	0.4224	0.0153	0.4115

Table 2. Meta-path study on the Amazon dataset.

Meta-path	Micro-F1	Macro-F1	Sample-F1
User-Product	0.8358	0.7306	0.8170
User-Persona	0.9305	0.8184	0.9249

Automatic Anchor Selection. We select optimal path P^* from a candidate set $\mathcal{P}_{\mathrm{cand}}$ via

$$P^* = \arg\max_{P \in \mathcal{P}_{\mathrm{cand}}} \frac{1}{2} \left(\text{Micro-F1}_{\mathrm{val}} + \text{Macro-F1}_{\mathrm{val}} \right) \quad (2)$$

Table 3. Performance of anchor selection protocol across datasets

Dataset	Selection	Micro-F1	Macro-F1	Sample-F1
IMDB	All	0.6511	0.6387	0.6420
	Manual	0.6692	0.6477	0.6513
	Automatic	0.6671	0.6415	0.6522
Amazom	All	0.8564	0.7852	0.8331
	Manual	0.9305	0.8184	0.9249
	Automatic	0.9305	0.8184	0.9249
CITE	All	0.3687	0.0095	0.3031
	Manual	0.4224	0.0153	0.4115
	Automatic	0.4006	0.0120	0.3994